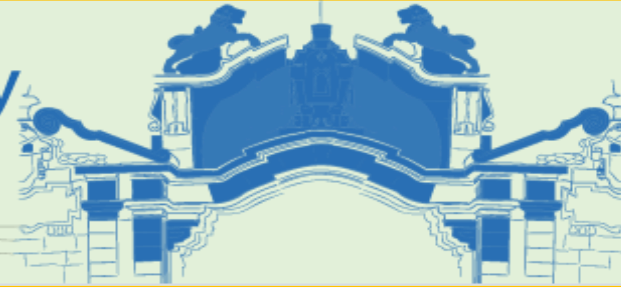




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# Addis Ababa Institute of Technology

School of Mechanical and Industrial Engineering

Engineering Materials I

(MEng 2122)

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*July, 2022*

Chapter - 7

## Dislocations and Strengthening Mechanisms

## Content Outline

- Dislocations
- Plastic deformation
- Slip system
- Stress and dislocation motion
- Mechanisms for strengthening

# Why we Study Dislocations and Strengthening Mechanisms?

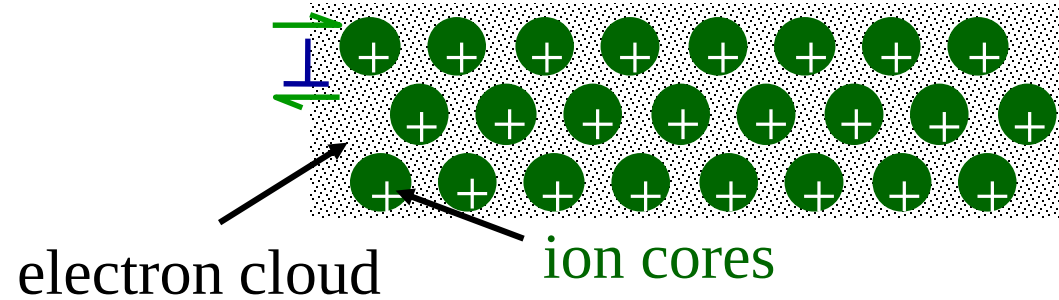
- Why are dislocations observed primarily in metals and alloys?
- How are strength and dislocation motion related?
- How do we increase strength?
- How can heating change strength and other properties?

# Dislocations and materials classes

## Metals:

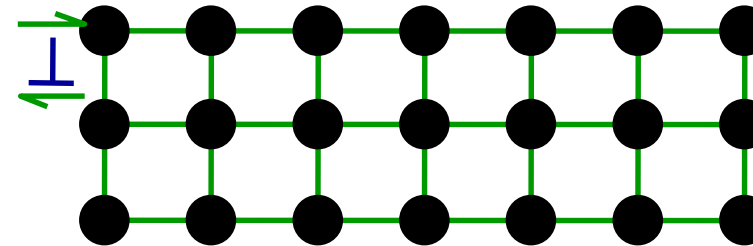
Dislocation motion easier.

- non-directional bonding
- close-packed directions for slip.



## Covalent Ceramics

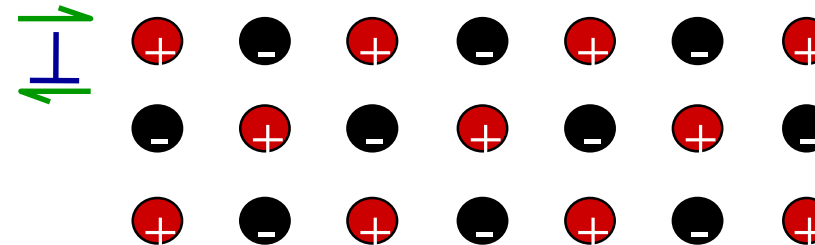
- (Si, diamond): Motion hard.
- directional (angular) bonding



- **Ionic Ceramics** (NaCl):

Motion hard.

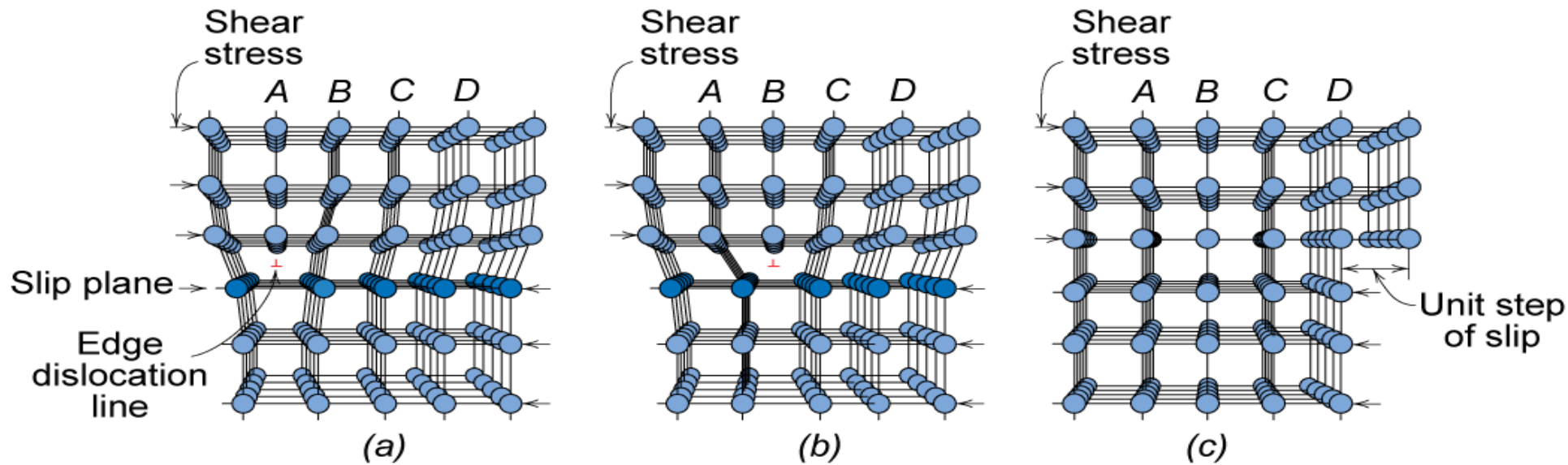
- need to avoid ++ and -- neighbors.



# Dislocations and Plastic Deformation

## Dislocations & plastic deformation:

Cubic and hexagonal metals - plastic deformation is by **plastic shear or slip** where one plane of atoms slides over adjacent plane by defect motion (dislocations).



# Dislocations and Plastic Deformation

## Slip System

- The process by which **plastic deformation** is produced by dislocation motion is termed **slip**.
- Ordinarily there is a preferred plane, and in that plane there are specific directions along which dislocation motion occurs. This plane is called the **slip plane**.

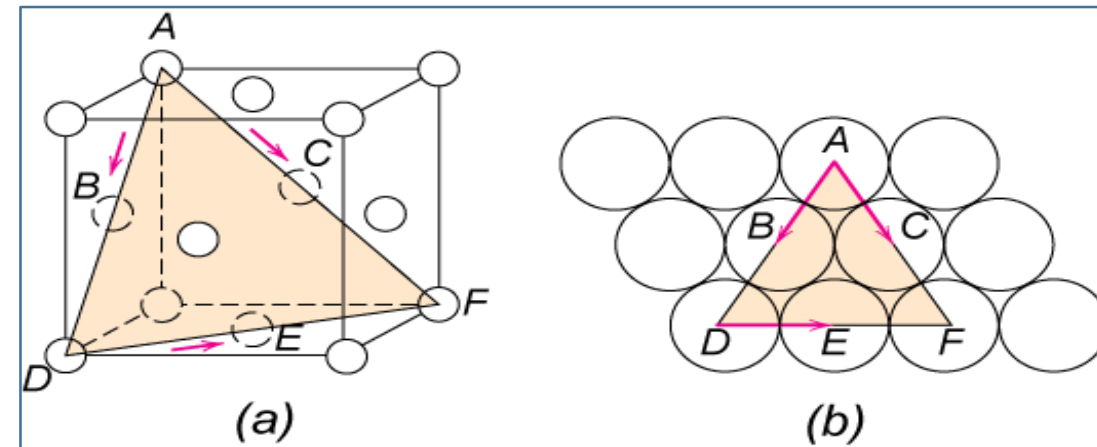
**Slip plane** - plane allowing easiest slippage

Wide interplanar spacings - highest planar densities

**Slip direction** - direction of movement - Highest linear densitie

FCC Slip occurs on  $\{111\}$  planes in  $\langle 110 \rangle$  directions  
=> total of 12 slip systems

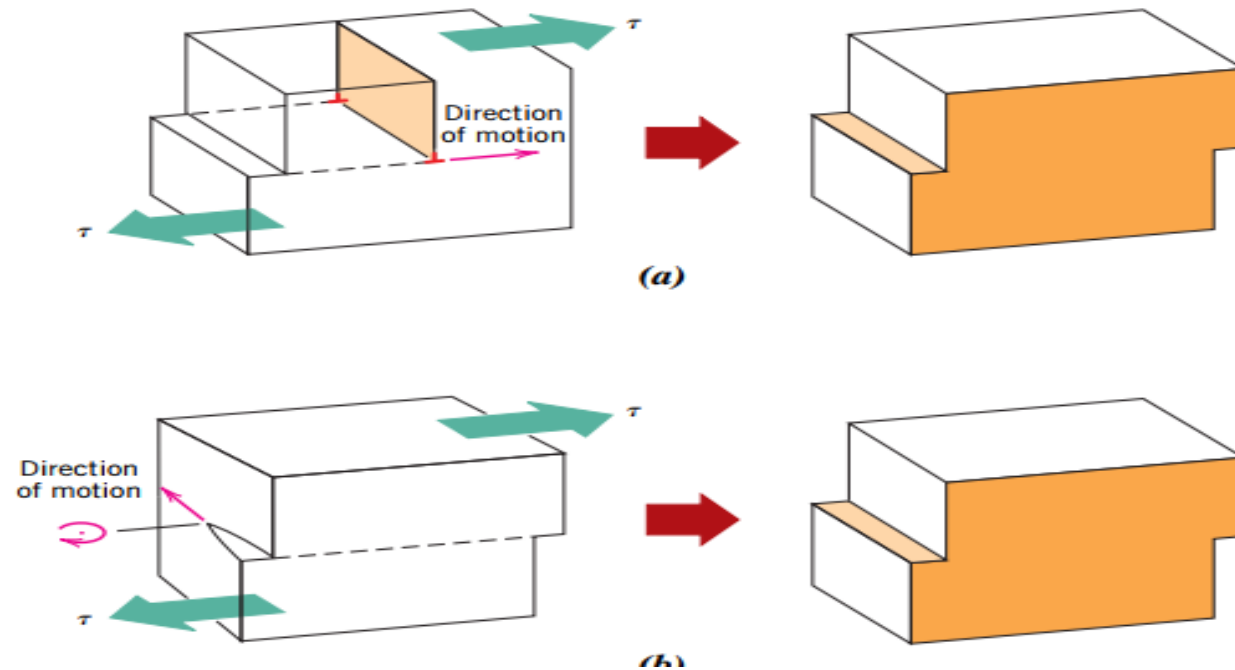
- in BCC & HCP other slip systems occur



- The combination of the slip plane and the slip direction is termed the **slip system**

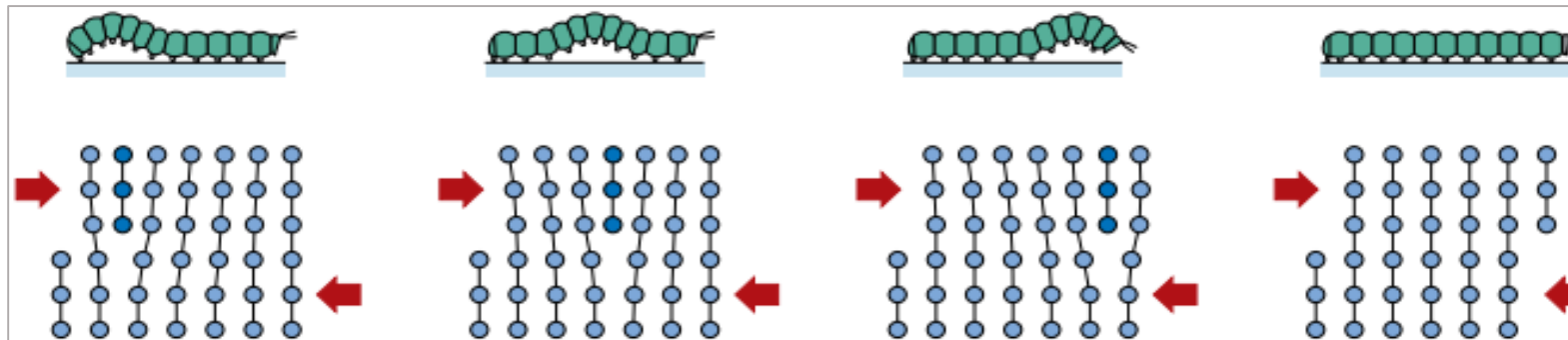
# Dislocations and Plastic Deformation

## Slip System



**Figure 7.2** The formation of a step on the surface of a crystal by the motion of (a) an edge dislocation and (b) a screw dislocation.

Note that for an **edge**, the dislocation line moves in the direction of the applied shear stress  $\tau$ ; for a **screw**, the dislocation line motion is perpendicular to the stress direction.



**Figure 7.3** The analogy between caterpillar and dislocation motion.

# Dislocations and Plastic Deformation

## Slip System

- All *metals* and *alloys* contain some dislocations that were introduced during solidification, during plastic deformation, and as a consequence of thermal stresses that result from rapid cooling.
- The number of dislocations, or **dislocation density** in a material, is expressed as the total *dislocation length* per *unit volume* or,
- equivalently, the **number of dislocations** that intersect a unit area of a random section.
- The units of **dislocation density** are millimeters of dislocation per cubic millimeter or just per square millimeter.

# Slip Systems in Cubic Metals

- The combination of the **slip plane** and the **slip direction** is termed the **slip system**.
- The **slip system** depends on the crystal structure of the material.

Table: Slip Systems for, Face-Centered Cubic, Body-Centered Cubic, and Hexagonal Close-Packed Metals.

| <i>Metals</i>                 | <i>Slip Plane</i> | <i>Slip Direction</i>        | <i>Number of Slip Systems</i> |
|-------------------------------|-------------------|------------------------------|-------------------------------|
| <b>Face-Centered Cubic</b>    |                   |                              |                               |
| Cu, Al, Ni, Ag, Au            | {111}             | $\langle 1\bar{1}0 \rangle$  | 12                            |
| <b>Body-Centered Cubic</b>    |                   |                              |                               |
| $\alpha$ -Fe, W, Mo           | {110}             | $\langle \bar{1}11 \rangle$  | 12                            |
| $\alpha$ -Fe, W               | {211}             | $\langle \bar{1}11 \rangle$  | 12                            |
| $\alpha$ -Fe, K               | {321}             | $\langle \bar{1}11 \rangle$  | 24                            |
| <b>Hexagonal Close-Packed</b> |                   |                              |                               |
| Cd, Zn, Mg, Ti, Be            | {0001}            | $\langle 11\bar{2}0 \rangle$ | 3                             |
| Ti, Mg, Zr                    | {10 $\bar{1}$ 0}  | $\langle 11\bar{2}0 \rangle$ | 3                             |
| Ti, Mg                        | {10 $\bar{1}$ 1}  | $\langle 11\bar{2}0 \rangle$ | 6                             |

- Burgers vectors for face-centered cubic, body-centered cubic, and hexagonal close packed crystal structures are given by

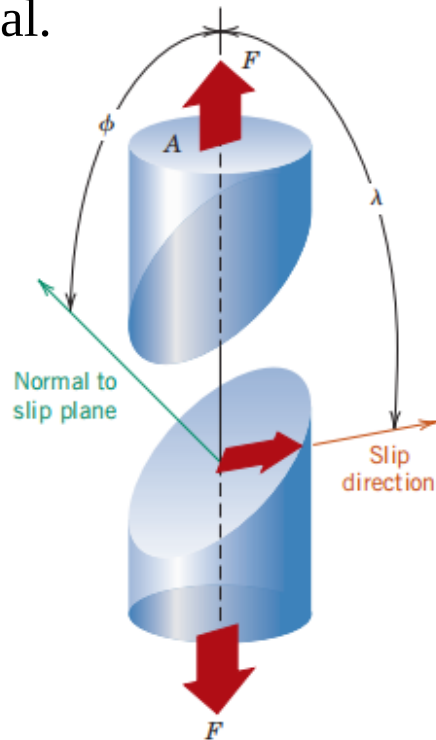
$$\mathbf{b}(\text{FCC}) = \frac{a}{2} \langle 110 \rangle$$

$$\mathbf{b}(\text{BCC}) = \frac{a}{2} \langle 111 \rangle$$

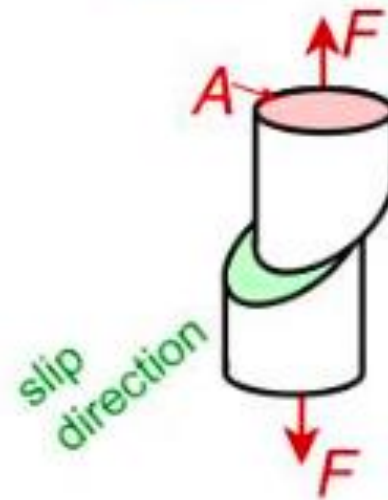
$$\mathbf{b}(\text{HCP}) = \frac{a}{3} \langle 11\bar{2}0 \rangle$$

# Slip in Single Crystals

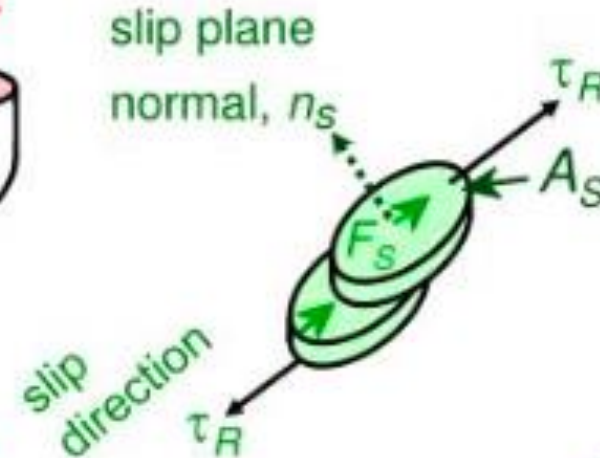
Geometrical relationships between the tensile axis, slip plane, and slip direction used in calculating the resolved shear stress for a single crystal.



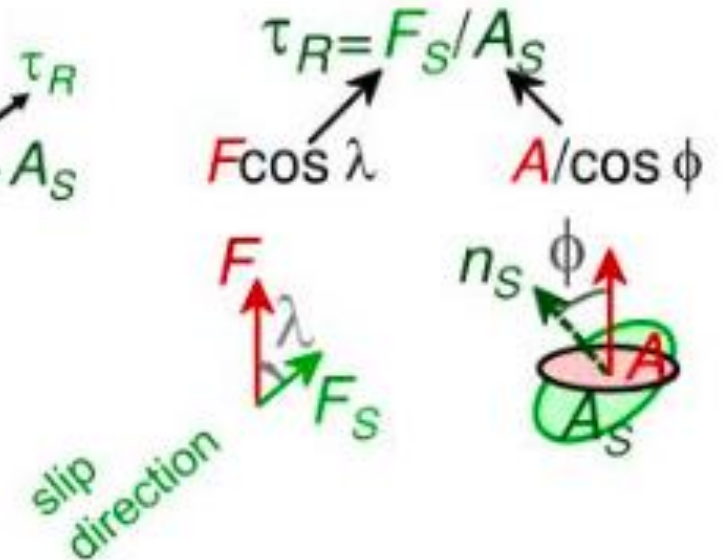
Applied tensile stress:  $\sigma = F/A$



Resolved shear stress:  $\tau_R = F_S/A_S$



Relation between  $\sigma$  and  $\tau_R$

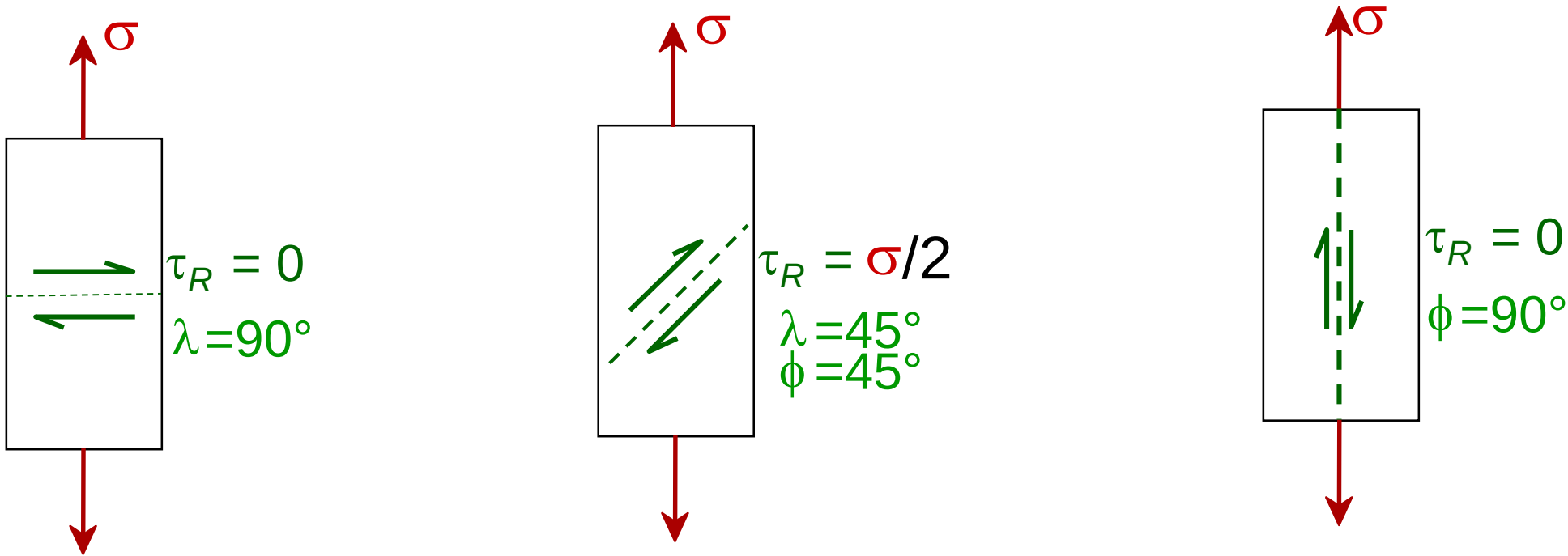


Crystals slip due to a resolved shear stress,  $\tau_R$

$$\tau_R = \sigma \cos \lambda \cos \phi$$

# Critical Resolved Shear Stress

- Condition for dislocation motion:  $\tau_R > \tau_{\text{CRSS}}$  typically  $10^{-4}$  GPa to  $10^{-2}$  GPa
- In response to an applied tensile or compressive stress, **slip in a single crystal commences** on the most favorably oriented slip system when the resolved shear stress reaches some critical value, termed the **critical resolved shear stress**.
- The minimum shear stress required to initiate slip and is a property of the material that determine when yielding occurs.  $\tau_R = \sigma \cos \lambda \cos \phi$
- Crystal orientation can make it easy or hard to move dislocation.



# Critical Resolved Shear Stress

$$\tau_{crss} = \sigma_y (\cos \lambda \cos \phi)_{\max}$$

$$\sigma_y = \frac{\tau_{crss}}{(\cos \lambda \cos \phi)_{\max}}$$

$\sigma_y$  – *yielding stress*

Resolved  $\tau$  (shear stress) is **maximum** at  $\lambda = \phi = 45^\circ$ , and

$\tau_{CRSS} = \sigma_y/2$  for dislocations to move (in single crystals)

- The **minimum stress** necessary to introduce yielding occurs when

$$\phi = \lambda = 45^\circ \text{ then}$$

$$\sigma_y = 2\tau_{crss}$$

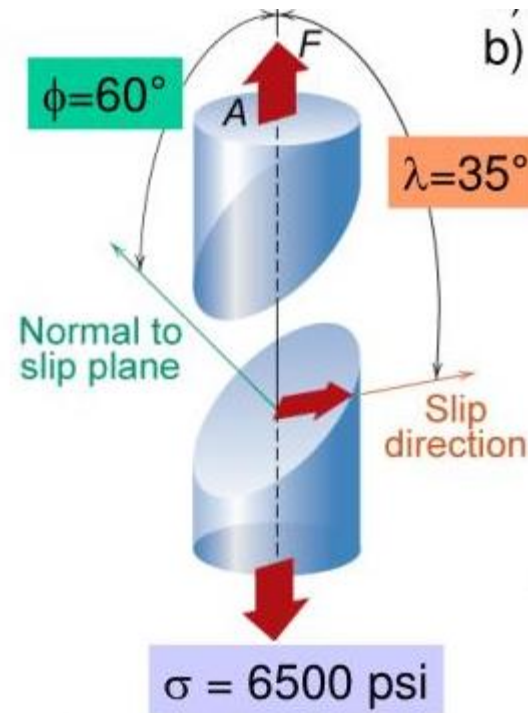
For  $\tau_{crss} = 0$

The crystal ordinarily fractures rather than deforming plastically

## Example: Deformation of single crystal

- a. will the single crystal Yield?
- b. if not what stress is needed?

$$\tau_{\text{crss}} = 3000 \text{ psi}$$



# Slip Motion in Polycrystals

- Stronger since grain boundaries pin deformations
- Slip planes & directions ( $\lambda$ ,  $\phi$ ) change from one crystal to another.
- $\tau_R$  will vary from one crystal to another.
- The crystal with the largest  $\tau_R$  yields first.
- Other (less favorably oriented) crystals yield (slip) later.



# Effect of Polycrystalline Materials

- *Ordinarily ductility* is sacrificed when an alloy is strengthened.
- The relationship between *dislocation motion* and *mechanical behavior* of metals is significant to the understanding of strengthening mechanisms.
- The *ability* of a metal to plastically deform depends on the ability of *dislocations to move*.
- *Restricting* or *hindering* dislocation motion renders a material harder and stronger.
- Consider strengthening *single phase metals* by: *grain size reduction*, *solid-solution alloying*, and *strain hardening*.
- *Deformation* and *slip* in *polycrystalline materials* is somewhat more complex.
- Because of the *random crystallographic* orientations of the numerous grains, the direction of slip varies from one grain to another.

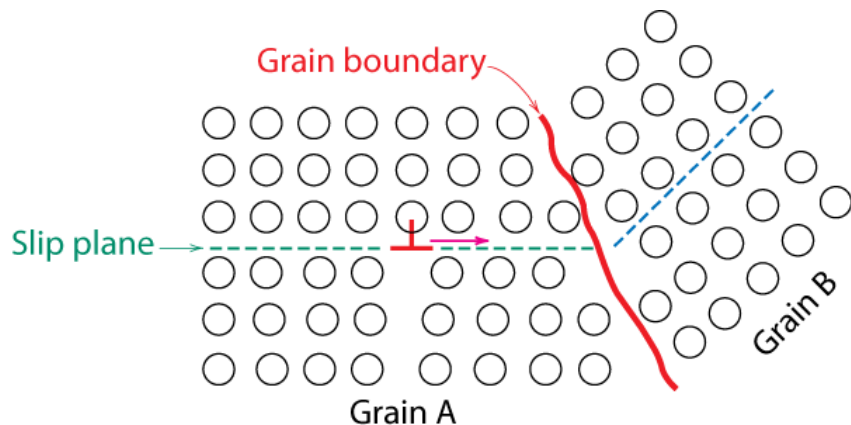
# Mechanisms for Strengthening

- The most common strengthening mechanisms in metals are;
  - 1) Reducing grain size - strengthening by grain refinement
  - 2) Solid solution – strengthening by alloying
  - 3) Precipitation hardening - age hardening
  - 4) Cold working - strain hardening
  - 5) Dispersion hardening
  - 6) Particulate strengthening
  - 7) Phase transformation hardening.

# Mechanisms for Strengthening

## 1) Reducing Grain Size:

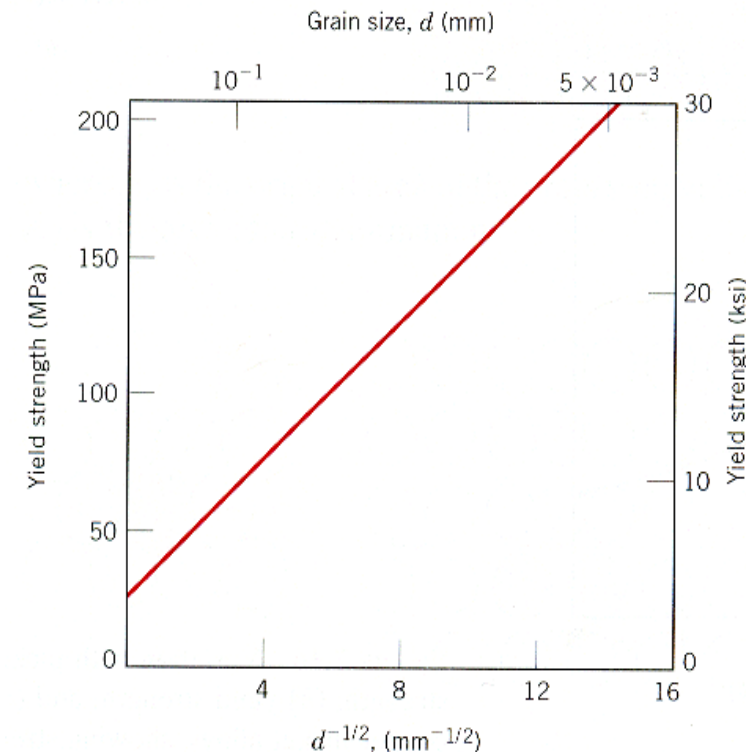
- Strength of a material is its resistance against deformation, especially, against plastic deformation or yielding.
- Grain boundaries are barriers to slip.
- Barrier "strength" increases with increasing angle of misorientation.
- Smaller grain size: more barriers to slip.



- Hall-Petch equation:

$$\sigma_{yield} = \sigma_o + k_y d^{-1/2}$$

Where,  $k$  is a constant,  $d$  is the average grain diameter and  $\sigma_o$  is the original yield stress.



**FIGURE 7.15** The influence of grain size on the yield strength of a 70 Cu-30 Zn brass alloy. Note that the grain diameter increases from right to left and is not linear. (Adapted from H. Suzuki, "The Relation Between the Structure and Mechanical Properties of Metals," Vol. II, *National Physical Laboratory, Symposium No. 15*, 1963, p. 524.)

# Mechanisms for Strengthening

## 1) Reducing Grain Size:

Grain size reduction techniques:

- *Increase* rate of solidification from the liquid phase.
- *Perform plastic deformation* followed by an appropriate heat treatment.
- Grain boundaries are regions where atoms are at higher energy level and also where atomic orientation changes.

Notes:

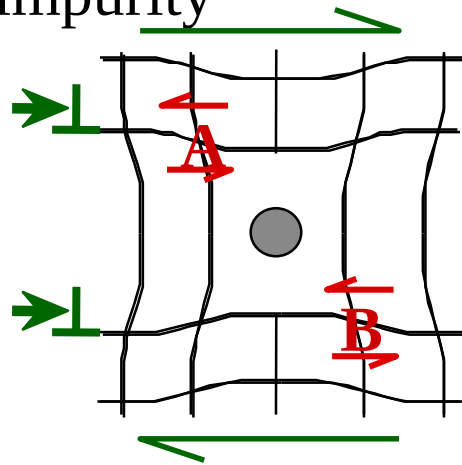
- *Grain size reduction* also improves toughness of many alloys.
- *Small-angle grain boundaries* are **not effective** in interfering with the slip process because of the small crystallographic misalignment across the boundary.
- *Boundaries* between *two different phases* are also impediments to movements of dislocations.

# Mechanisms for Strengthening

## 2) Solid Solutions:

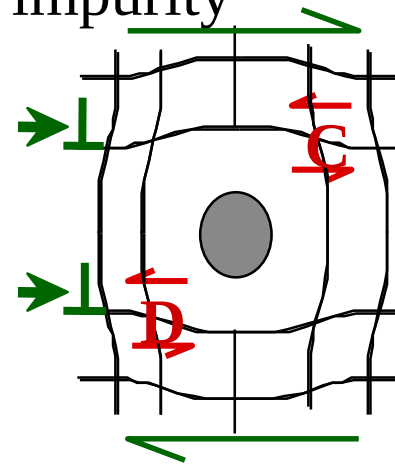
- *Impurity atoms* distort the lattice and generate stress.
- *Stress* can produce a barrier to *dislocation motion*.

- Smaller substitutional impurity



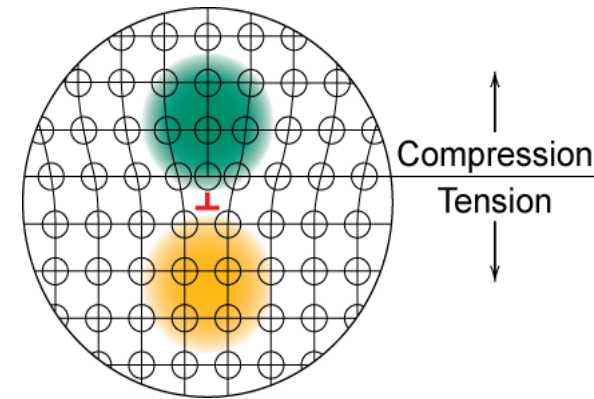
*Impurity generates local stress at A and B that opposes dislocation motion to the right.*

- Larger substitutional impurity



*Impurity generates local stress at C and D that opposes dislocation motion to the right.*

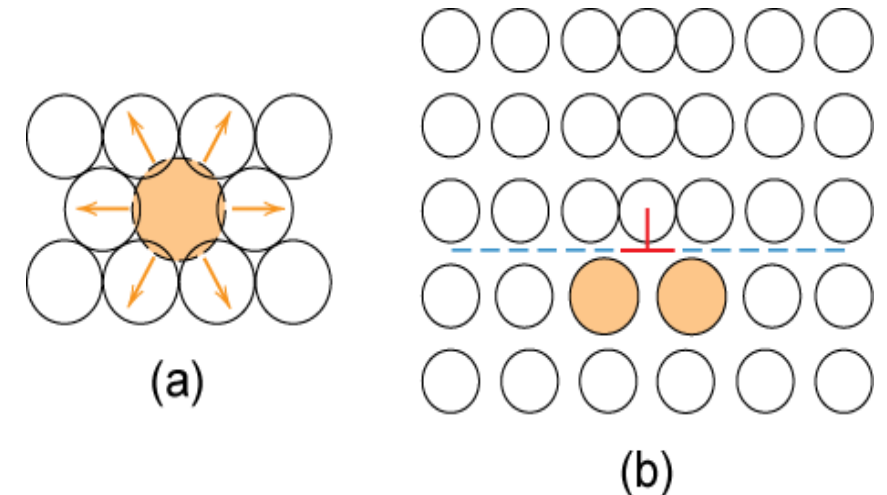
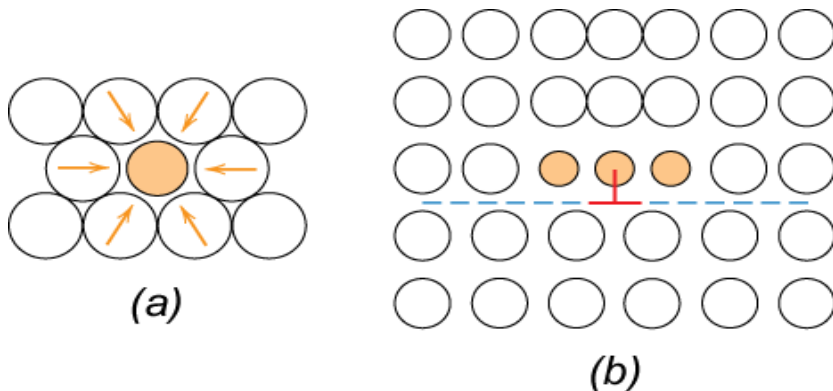
- Stress concentration at dislocations



# Mechanisms for Strengthening

## 2) Solid Solutions:

- Strengthening by Alloying
  - *Small impurities* tend to concentrate at *dislocations* on the “Compressive stress side”.
  - *Reduce mobility of dislocation* - increase strength.
- *Large impurities* concentrate at dislocations on “Tensile Stress” side – pinning dislocation.

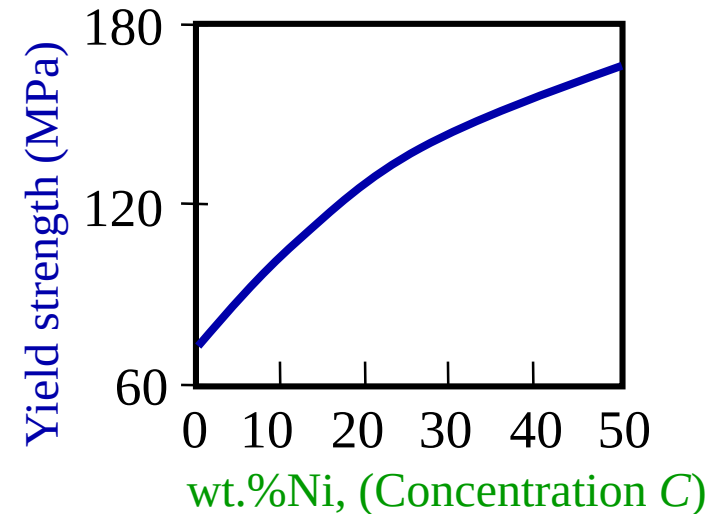
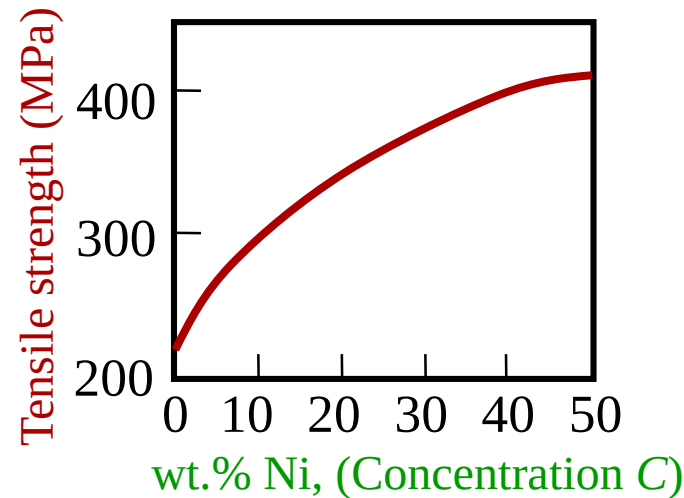


# Mechanisms for Strengthening

## 2) Solid Solutions:

Ex: Solid solution strengthening in copper

- Tensile strength and yield strength increase with wt% Ni.



- Empirical relation:  $\sigma_y \sim C^{1/2}$
- Alloying increases  $\sigma_y$  and  $TS$ .

# Mechanisms for Strengthening

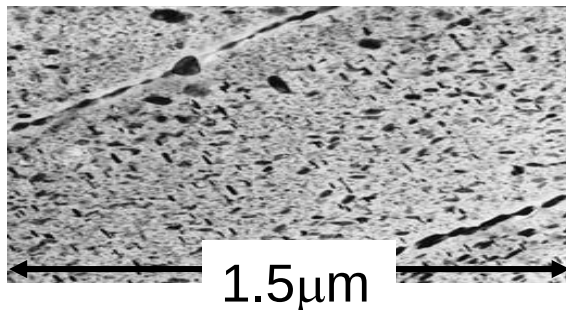
## 3) Precipitation Hardening:

Application of *precipitation strengthening*:

- *Internal wing structure* on Boeing aircraft.



- Aluminum is *strengthened with precipitates* formed by alloying and heat-treatment.



- *Hard precipitates* are difficult to shear. *Ex: Ceramics in metals (SiC in Iron or Aluminum).*

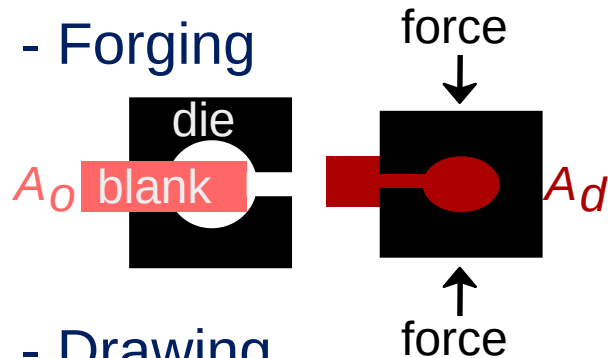
- The general steps involved in age/precipitation hardening are:
  - (i) Heating (solutionizing),
  - (ii) Quenching, and
  - (iii) Ageing /waiting

# Mechanisms for Strengthening

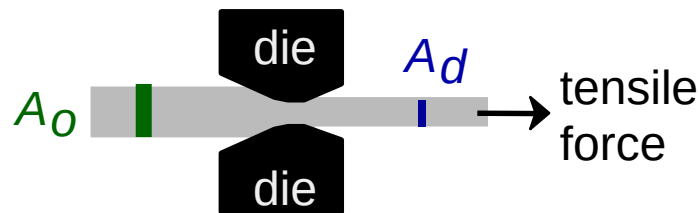
## 4) Cold Working:

- Room temperature deformation.
- Common *forming operations* change the *cross sectional area*:

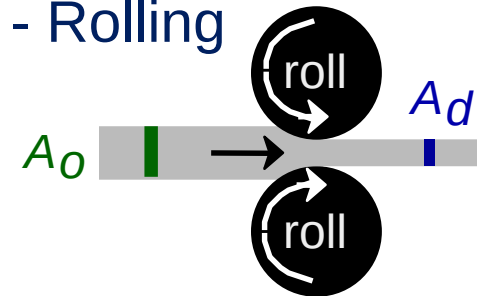
- Forging



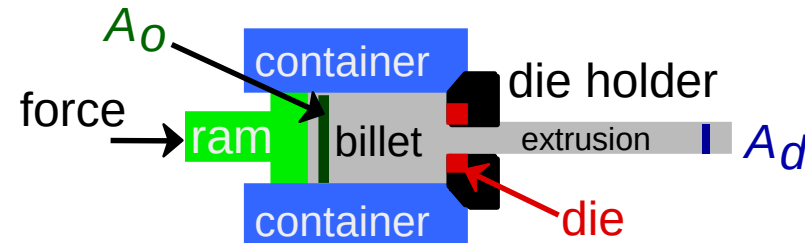
- Drawing



- Rolling



- Extrusion



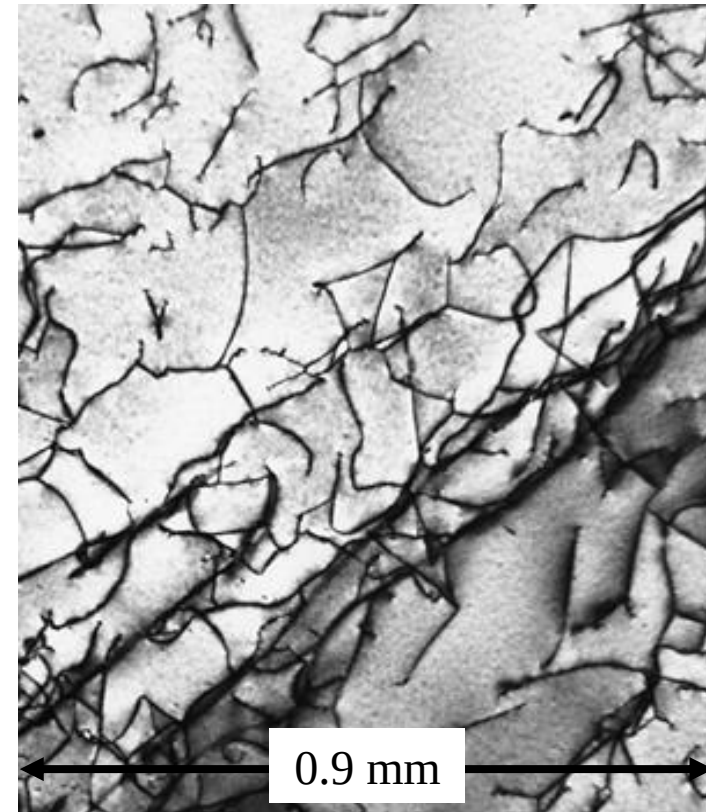
$$\%CW = \frac{A_o - A_d}{A_o} \times 100$$

# Mechanisms for Strengthening

## 4) Cold Working:

During cold work;

- Ti alloy after cold working:
- Dislocations entangle and multiply
- Thus, dislocation motion becomes more difficult.



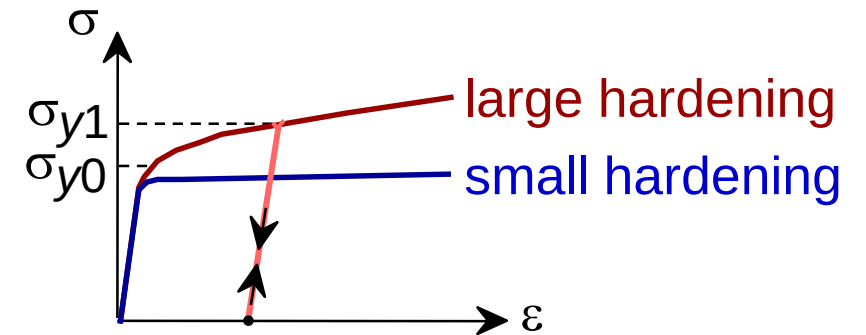
# Mechanisms for Strengthening

## 4) Cold Working:

### Result of cold work:

- Dislocation density =  $\frac{\text{total dislocation length}}{\text{unit volume}}$
- Carefully grown single crystal.
- Deforming sample increases density.
- Heat treatment reduces density.

- Yield stress increases as  $\rho_d$  increases:



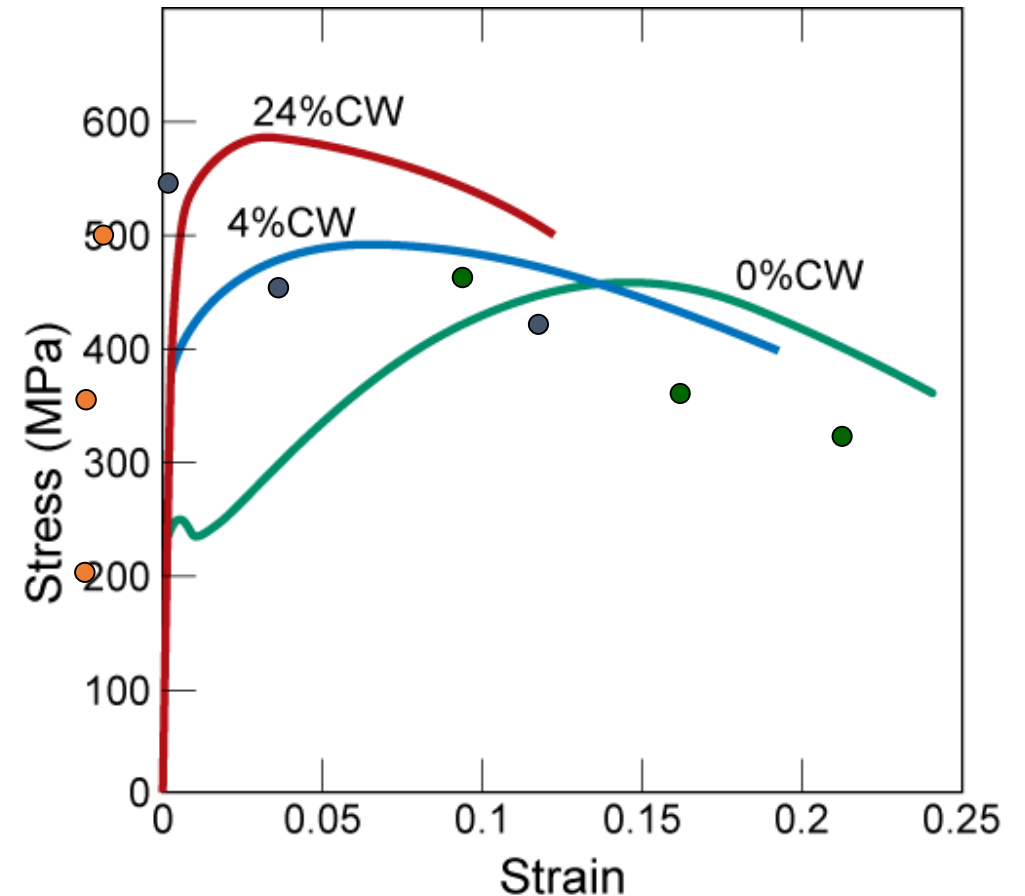
# Mechanisms for Strengthening

## 4) Cold Working:

### Impact of cold work:

As cold work is increased;

- Yield strength ( $\sigma_y$ ) increases.
- Tensile strength ( $TS$ ) increases.
- Ductility ( $\%EL$  or  $\%AR$ ) decreases.

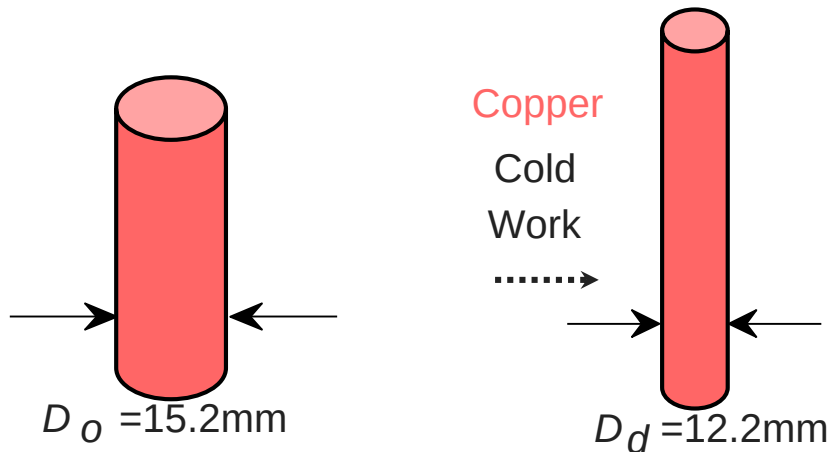


# Mechanisms for Strengthening

## 4) Cold Working:

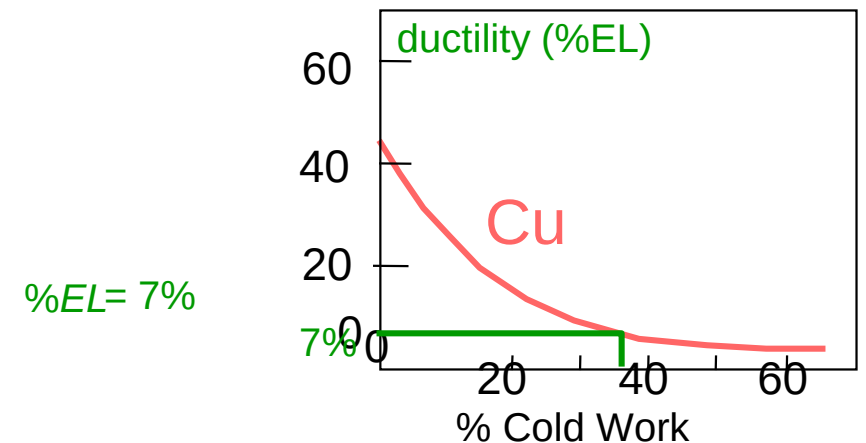
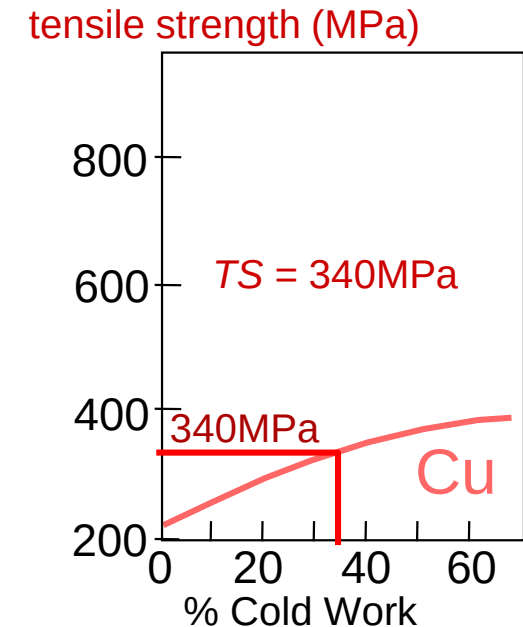
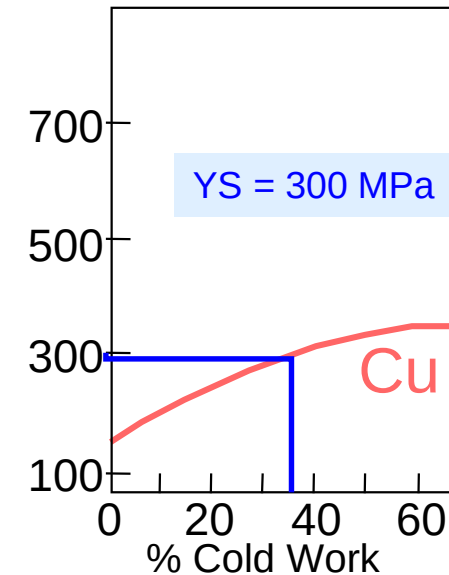
### Cold work analysis:

- What is the *TS* and ductility after cold working?



$$\%CW = \frac{\pi r_o^2 - \pi r_d^2}{\pi r_o^2} \times 100 = 35.6\%$$

- What is the *TS* and ductility after cold working to 35.6%?

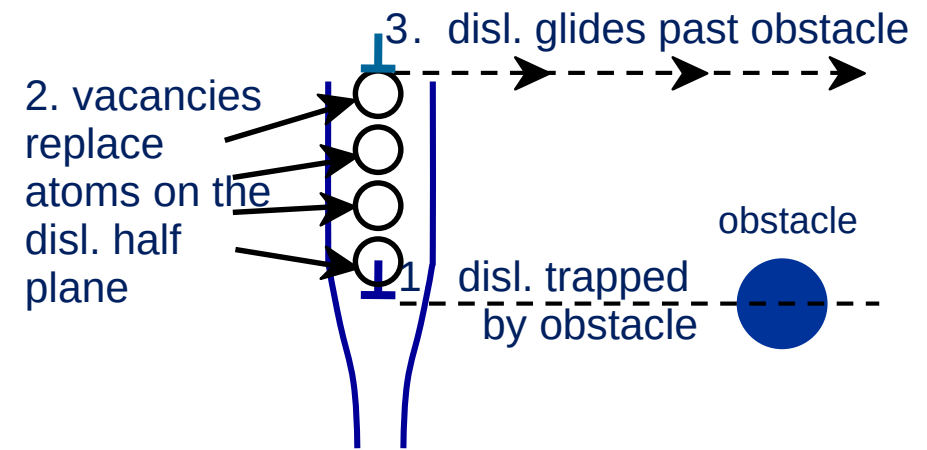
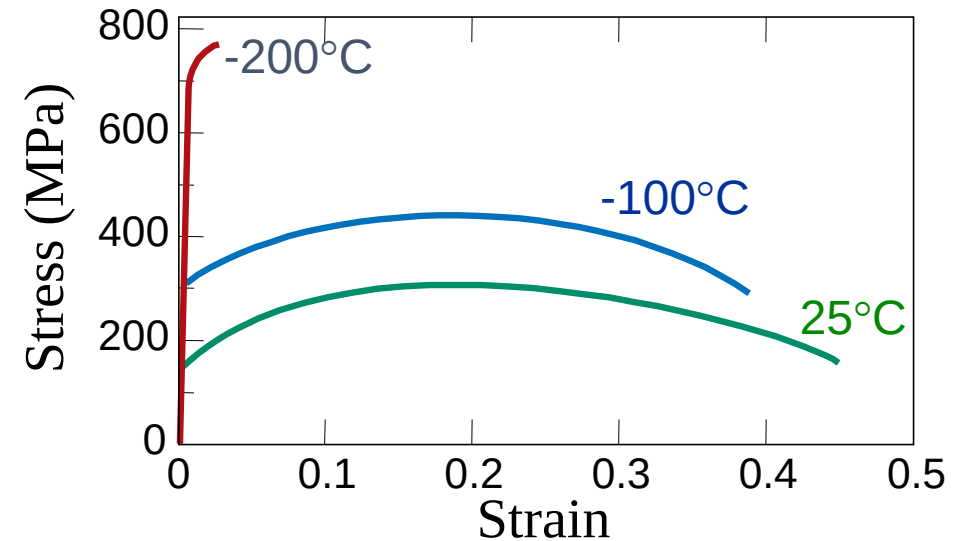


# Mechanisms for Strengthening

## 4) Cold Working:

- Results for polycrystalline iron:
  - $\sigma_y$  and  $TS$  *decrease* with increasing test temperature.
  - % $EL$  *increases* with increasing test temperature.
  - Why Vacancies help dislocations move past obstacles.

$\sigma$ -  $\epsilon$  Behavior vs. Temperature

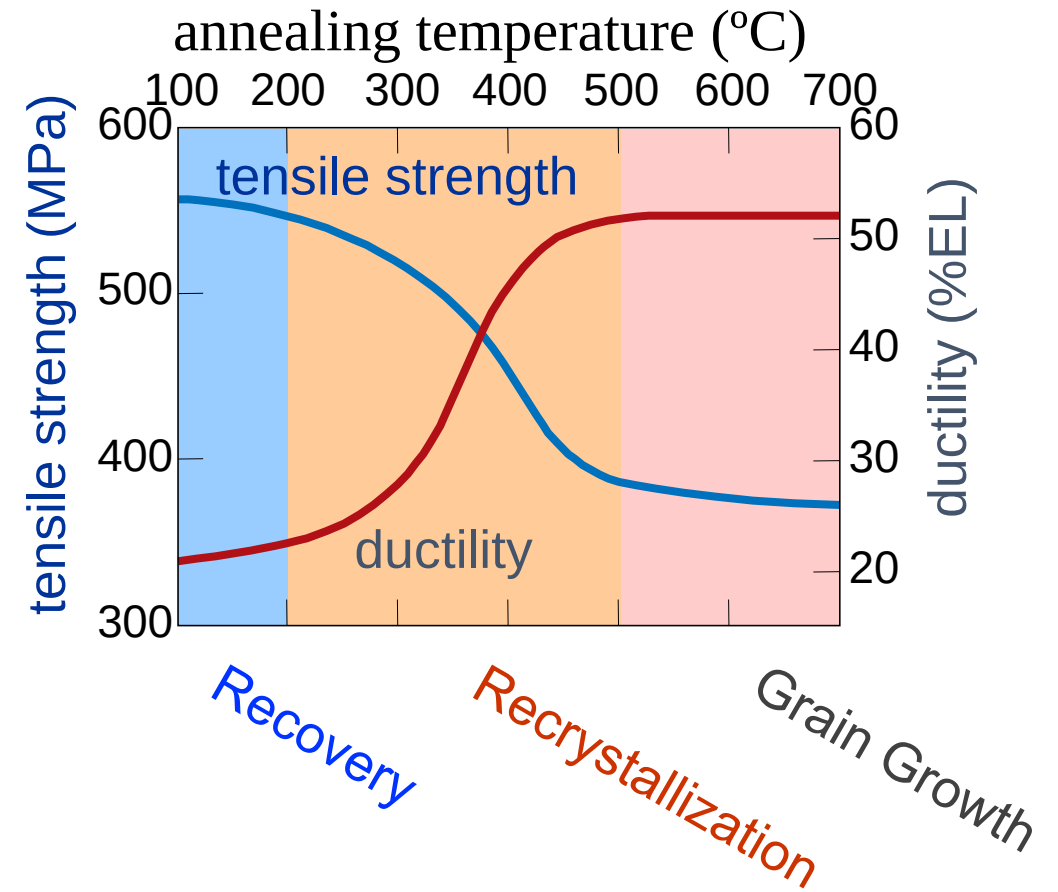


# Mechanisms for Strengthening

## 4) Cold Working:

### Effect of heating after %CW

- 1 hour treatment at  $T_{annealing}$   
decreases  $TS$  and increases % $EL$ .
- Effects of cold work are reversed!
- Three annealing stages;
  - 1) *Recovery*
  - 2) *Recrystallization*
  - 3) *Grain growth*

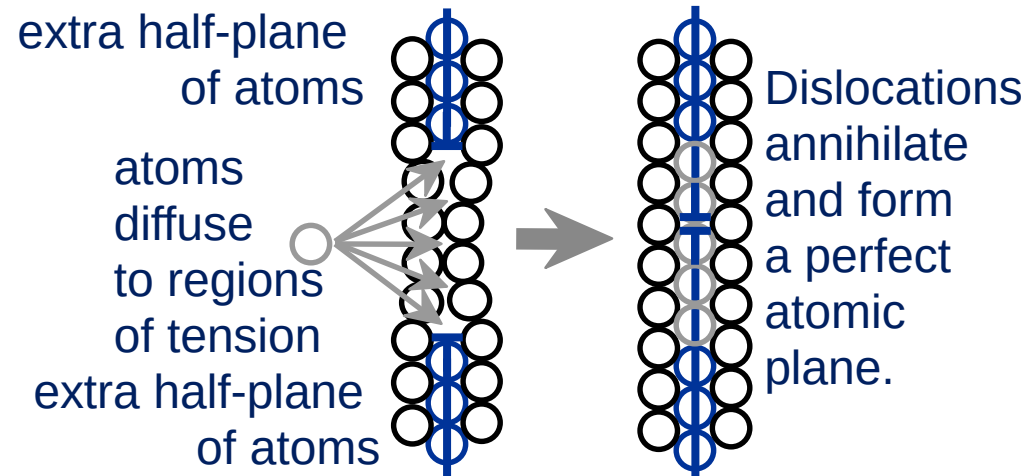


# Other Mechanisms for Strengthening

## Recovery:

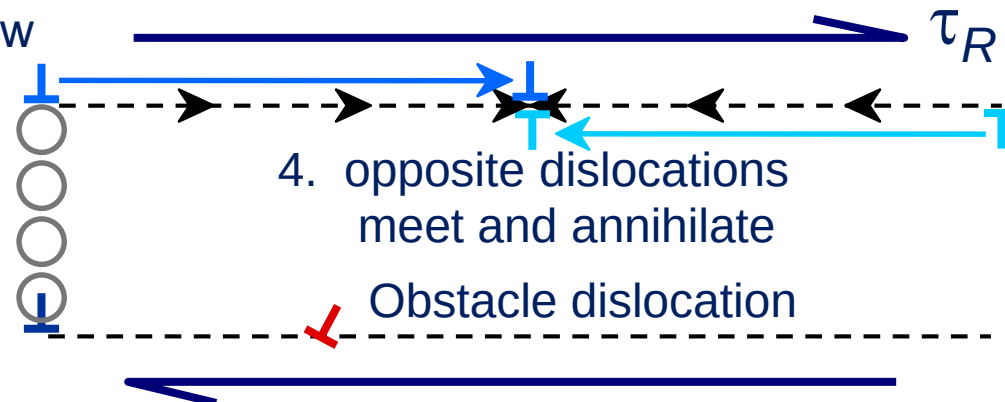
Annihilation reduces dislocation density.

- **Scenario 1**  
Results from diffusion



- **Scenario 2**

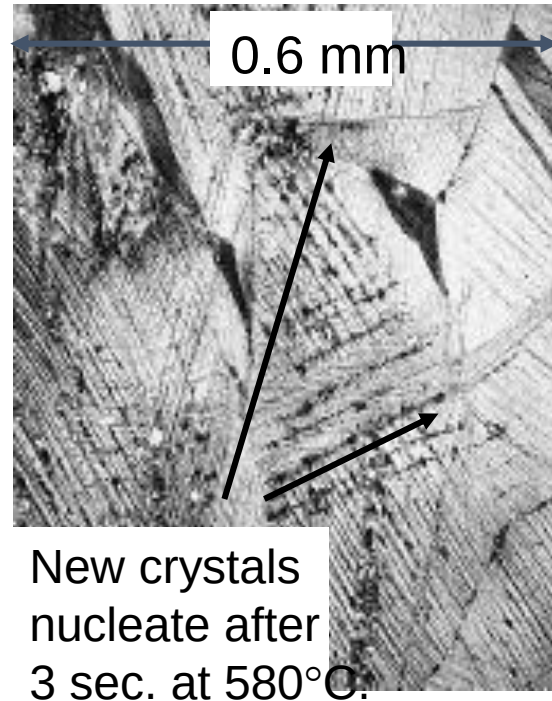
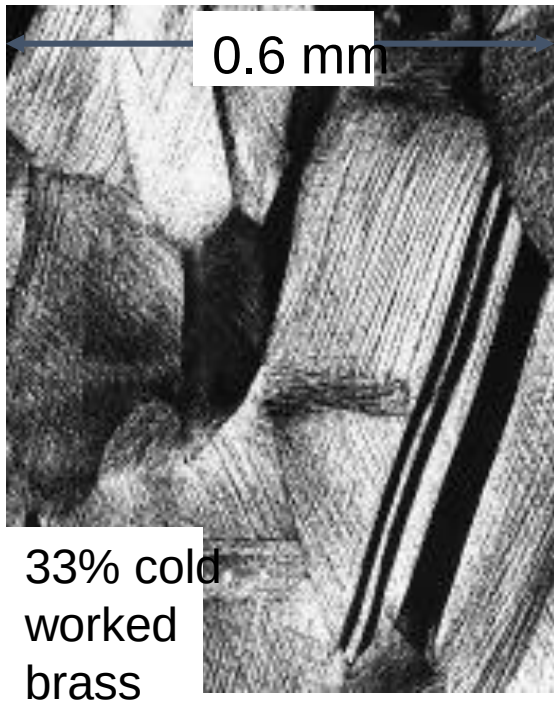
3. "Climbed" disl. can now move on new slip plane
2. grey atoms leave by vacancy diffusion allowing disl. to "climb"
1. dislocation blocked; can't move to the right



# Other Mechanisms for Strengthening

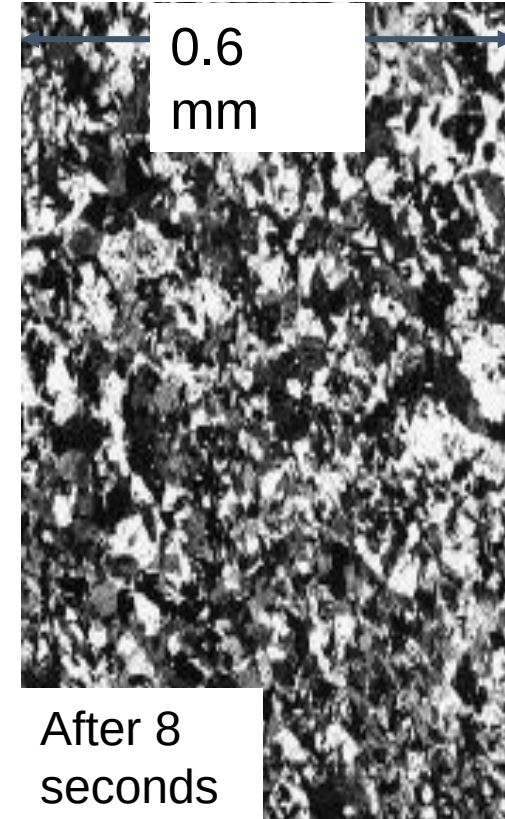
## Recrystallization:

- New grains are formed that:
  - have a low dislocation density
  - are small
  - consume cold-worked grains.



## Further Recrystallization

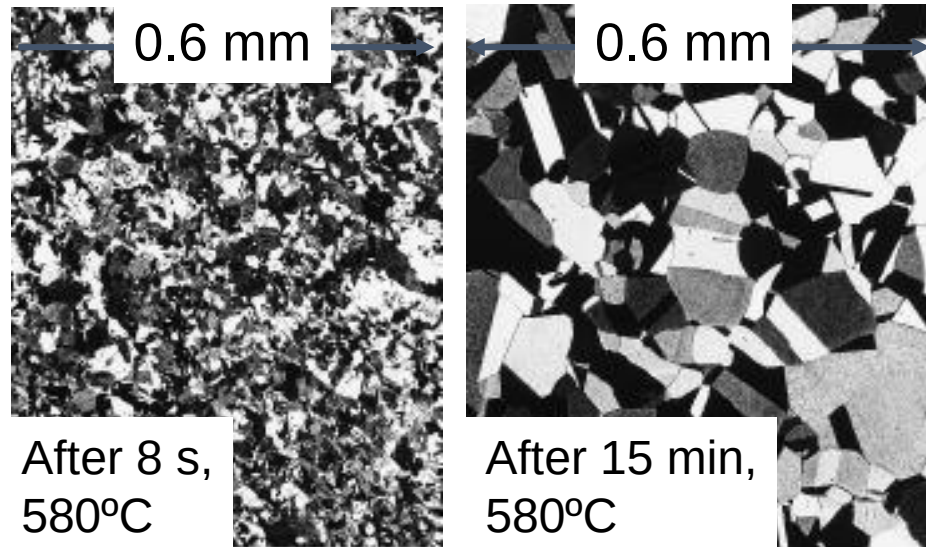
- All cold-worked grains are consumed.



# Other Mechanisms for Strengthening

## Grain Growth

- At longer times, larger grains consume smaller ones.
- Why?
- Grain boundary area (and therefore energy) is reduced.



- Empirical Relation:

Exponent type  $\sim 2$

$$d^n - d_o^n = Kt$$

Grain diameter at time t.

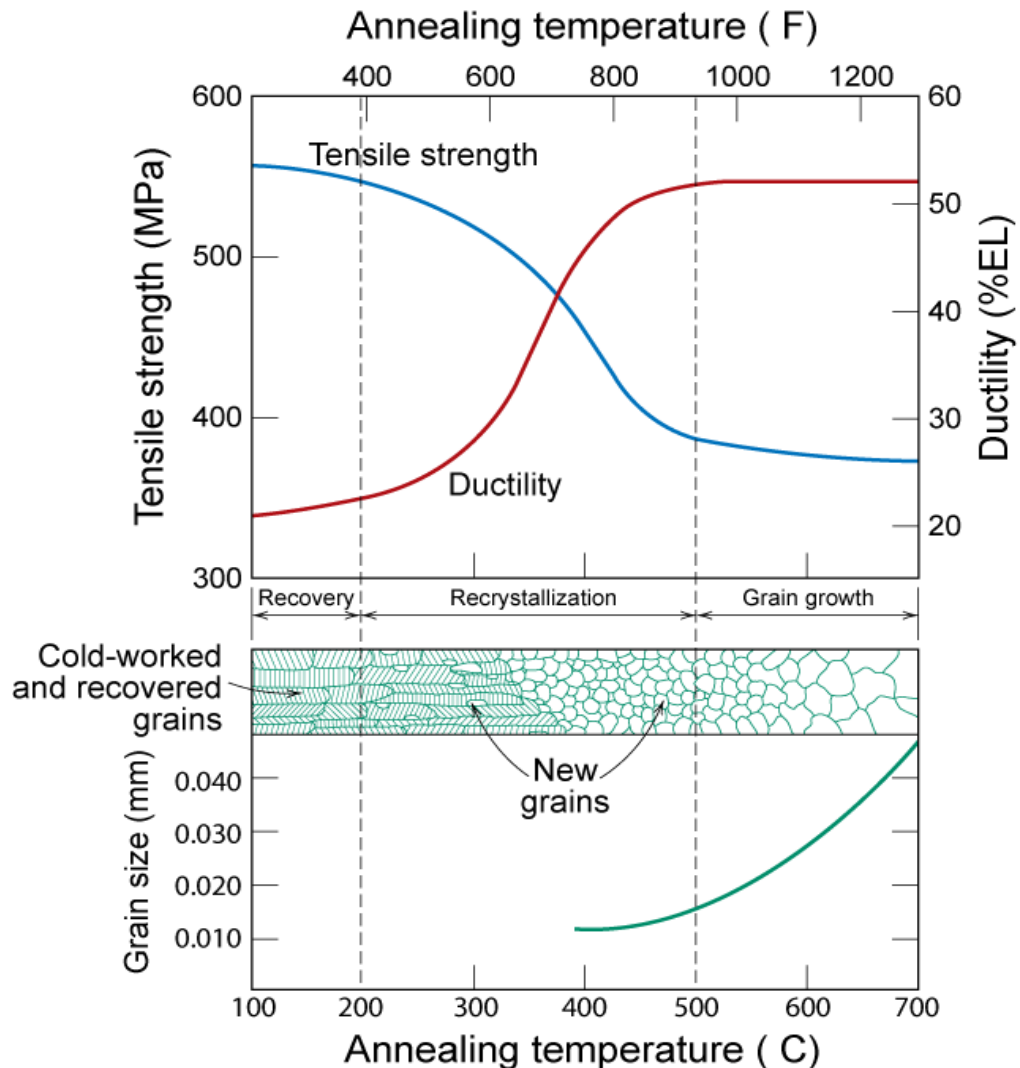
Coefficient dependent on material & Temp.

elapsed time

This is: Ostwald Ripening

# Other Mechanisms for Strengthening

## Grain Growth



$T_R$  = recrystallization temperature

$T_R$  = recrystallization temperature = point of highest rate of property change.

1.  $T_R \approx 0.3-0.6 T_m$  (K)
2. Due to diffusion  $\rightarrow$  annealing time  $\rightarrow T_R = f(t)$   
shorter annealing time  $\Rightarrow$  higher  $T_R$
3. Higher %CW  $\Rightarrow$  lower  $T_R$  – strain hardening
4. Pure metals lower  $T_R$  due to dislocation movements  
easier to move in pure metals  $\Rightarrow$  lower  $T_R$ .

# Summary

- **Dislocations** are observed primarily in metals and alloys.
- **Strength** is increased by making dislocation motion difficult.
- Particular ways to **increase strength** are to:
  - decrease grain size
  - solid solution strengthening
  - precipitate strengthening
  - cold work
- **Heating** (annealing) can reduce dislocation density and increase grain size. This decreases the strength.

*End of the chapter!*

*Thank you!*